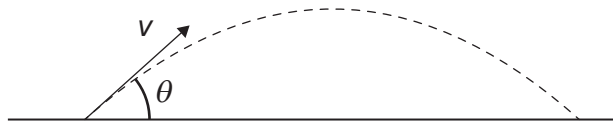


PAPER 1 / QUESTION 30

5 November 2025 / TZ1 (file label) / HL Paper 1A / original question 1 / 8825-6707

30. A projectile is launched with an initial velocity v at an angle θ above the horizontal. The projectile lands at the same height from which it was released. Air resistance is negligible.



What is the time of flight of the projectile?

- A. $\frac{v \sin \theta}{g}$
- B. $\frac{2v \sin \theta}{g}$
- C. $\frac{2v \cos \theta}{g}$
- D. $\frac{2v \sin \theta \cos \theta}{g}$